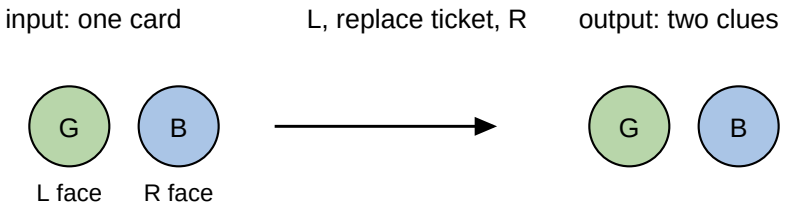
Choose RR, RB, or BB with equal chances; keep that card hidden. For each clue, independently draw an equal-chance L/R ticket, show that face, then replace and mix. Hide the ticket, face marks, and turning. The guesser sees only the two color clues.



Problem 1: Two clues are red, then red. Which card is the best guess? What if the clues are red, then blue? Settle both questions using all twelve equally likely card-and-face stories.

card	L then L	L then R	R then L	R then R
R R				
R B				
B B				

Fix the first choice at door 1. Choose the prize uniformly from doors 1, 2, 3. Independently draw ticket 2 or 3 with equal chances. Host A knows the prize and opens an unchosen empty door; use the ticket only if both are empty. Host B opens the ticket's door blindly; discard prize reveals. Hide prize placement and all tickets. The guesser sees only which empty door opens.

Same input: choose door 1; prize at door 2; ticket 2.

Host A: ignores ticket; opens 3

1 chosen	2 prize	3 open empty

Switch 1 to 2: win.

Host B: uses ticket; opens 2

1 chosen	2 open prize	3

Prize revealed: discard trial.

Problem 2: After an empty door opens, should you stay at door 1 or switch to the other closed door? Decide for each host using every prize-ticket story. Does the host's knowledge change the answer?

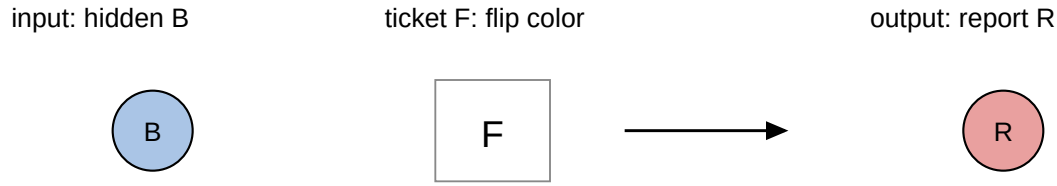
Host A

prize	ticket	opens	switch wins?
1	2		
1	3		
2	2		
2	3		
3	2		
3	3		

Host B

prize	ticket	opens	switch wins?
1	2		
1	3		
2	2		
2	3		
3	2		
3	3		

Draw a hidden counter uniformly from R1, R2, R3, B. The reporter sees its color, then independently draws H1, H2, or F with equal chances. H1 and H2 report the true color; F reports the other color. Hide the counter identity and ticket; show only the R/B report. Return and mix both draws each round.



Problem 3: A blue report arrives. Which hidden color is the better guess? Use every counter-and-report-ticket story. Replace the reporter's tickets with four equal-chance tickets, each honest or flipping. Can you make a blue report give both hidden colors equal chances?

counter	H1	H2	F
R1			
R2			
R3			
B			

1	2	3	4